# PROBLEM SOLVING USING PATTERN RECOGNITION
# DAY 1A

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# DAY 1 AGENDA

1.1 Introduction to Problem Solving Using Pattern Recognition

1.2 How to analyse, model and solve pattern recognition problems

1.3 Solving Pattern Recognition Problems Using Supervised Learning Techniques (I)

1.4 Pattern Recognition Workshop 1

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# 1.1
# Introduction to Problem Solving Using Pattern Recognition

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# Topics

- What is Pattern?
- What is Pattern Recognition?
- Examples of Pattern Recognition
- Applications of Pattern Recognition

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# Patterns

- Pattern is in everything around us in this digital world.

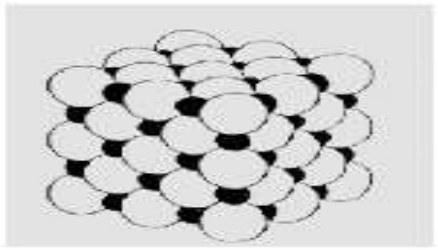
- Visual, audio, text, temporal…

- A pattern can either be seen physically or it can be observed mathematically by applying algorithms.

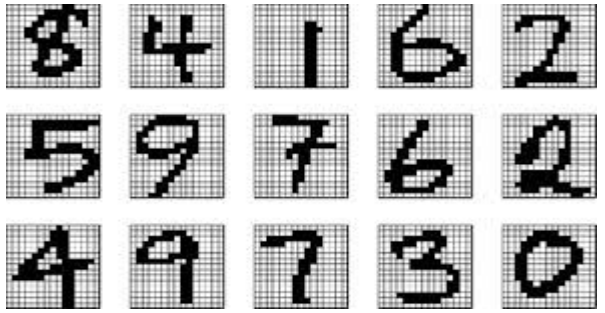
  e.g.: The colours on the clothes, speech pattern etc.

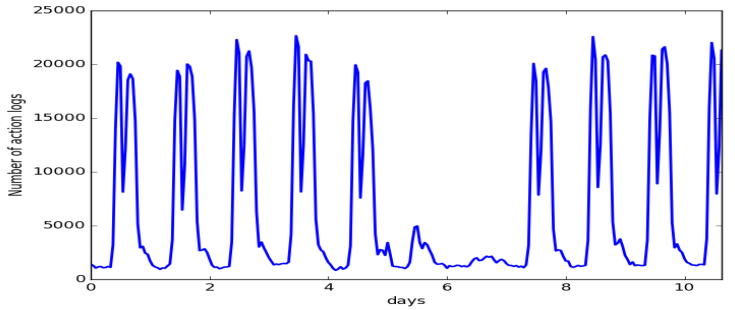
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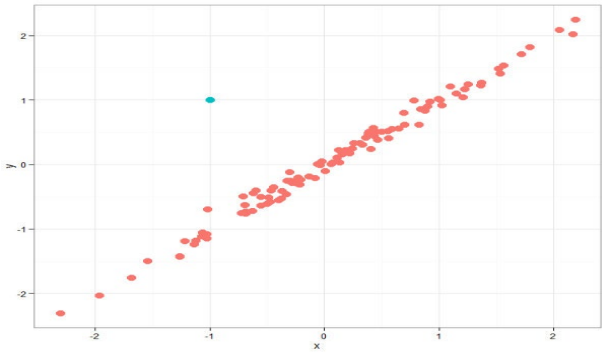
# Patterns

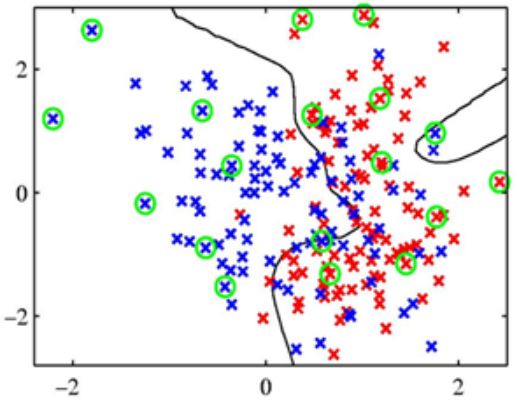


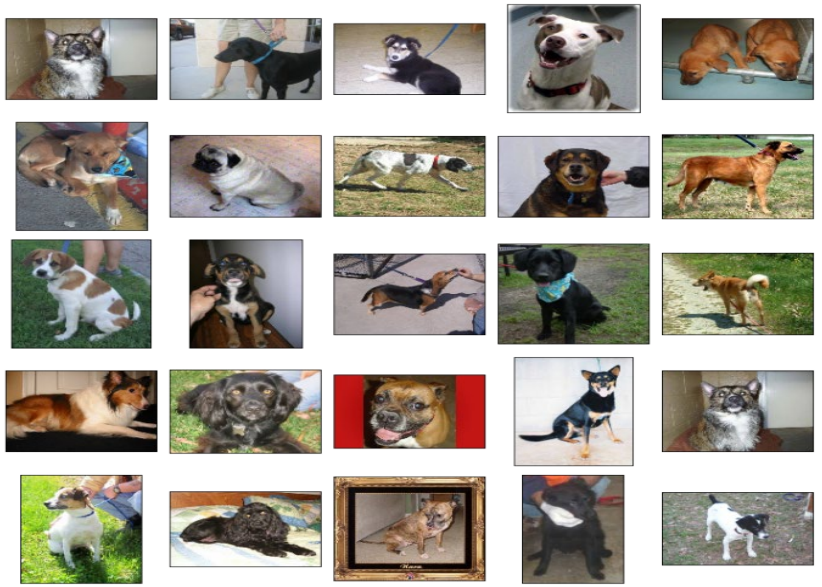








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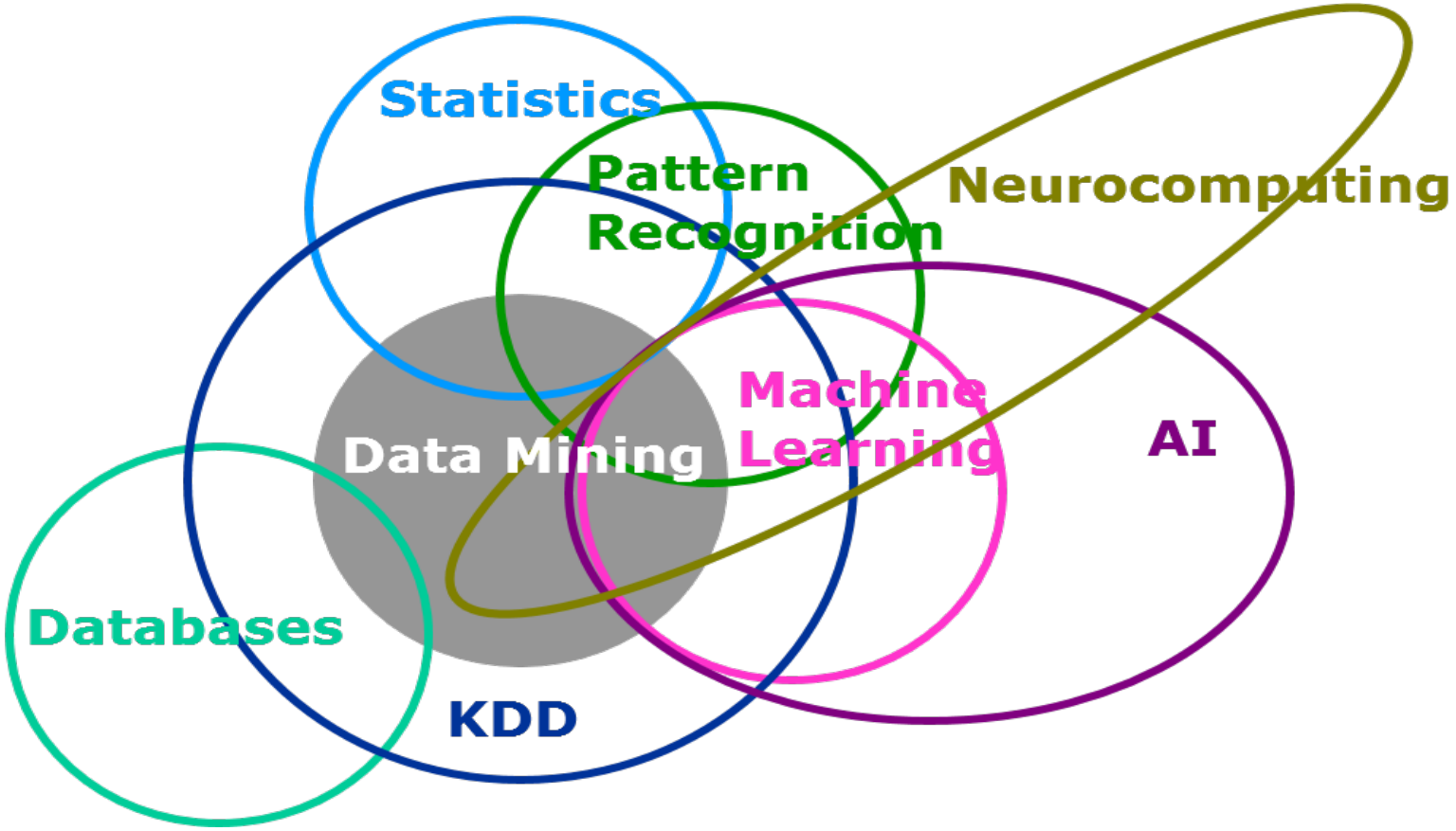
# Pattern Recognition

- Pattern recognition is the automated recognition of patterns and regularities in data. Pattern recognition is closely related to artificial intelligence and machine learning, together with applications such as data mining and knowledge discovery in databases, and is often used interchangeably with these terms.

--- Wikipedia

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# Pattern Recognition



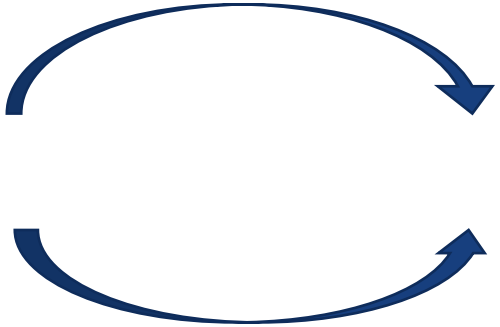

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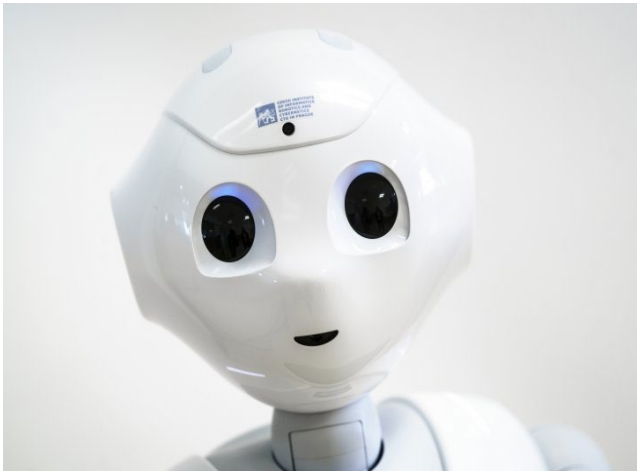
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# Pattern Recognition

## From Human Perception to Machine Perception







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# From Human Perception to Machine Perception

- **Humans have developed highly sophisticated skills for sensing their environment and taking actions according to what they observe, e.g.,**
  - Recognizing a face.
  - Understanding spoken words.
  - Reading handwriting.
  - Distinguishing fresh food from its smell.
- **We are often influenced by the knowledge of how patterns are modeled and recognized in nature when we develop pattern recognition algorithms.**
- **Research on machine perception also helps us gain deeper understanding and appreciation for pattern recognition systems in nature.**

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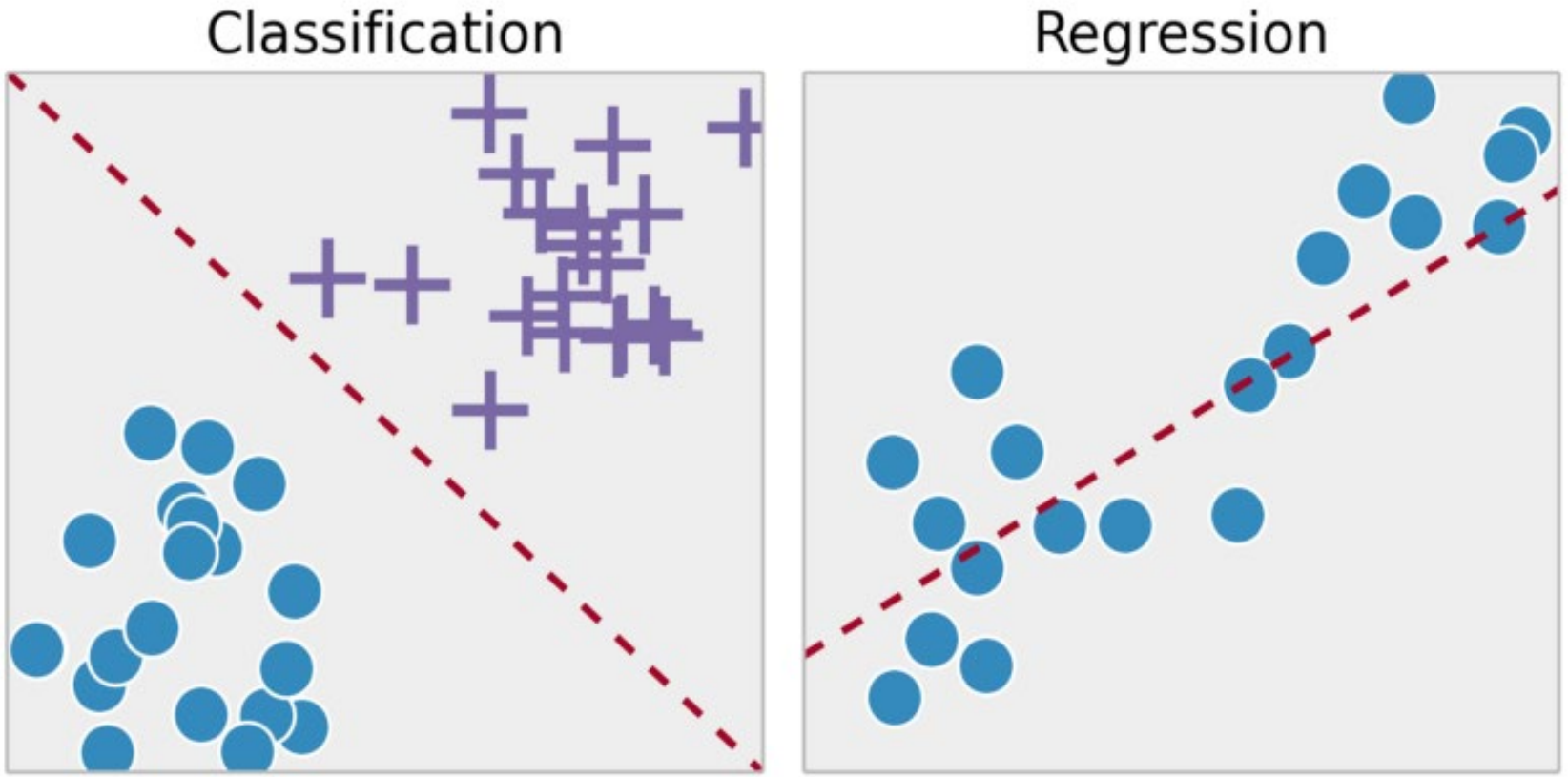
# Machine Leaning for PR

"Pattern Recognition is a type of problem while Machine Learning is a type of solution."

- **Supervised learning**
  - A teacher provides a category label or cost for each pattern in the training set.
- **Unsupervised learning**
  - The system forms clusters or natural groupings of the input patterns (based on some similarity criteria).
- **Reinforcement learning**
  - No desired category is given but the teacher provides feedback to the system such as the decision is right or wrong.

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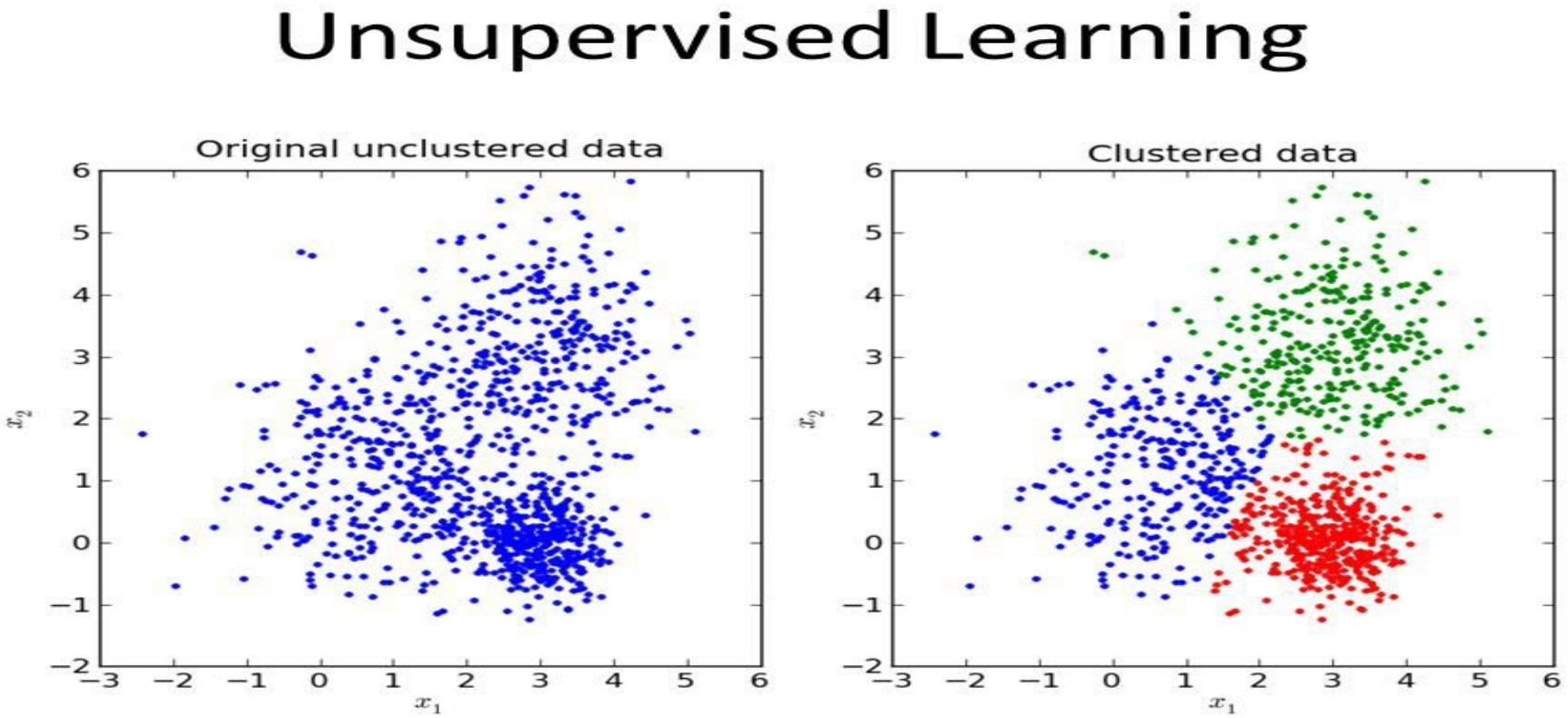
# Supervised Learning




Source: KDnuggets

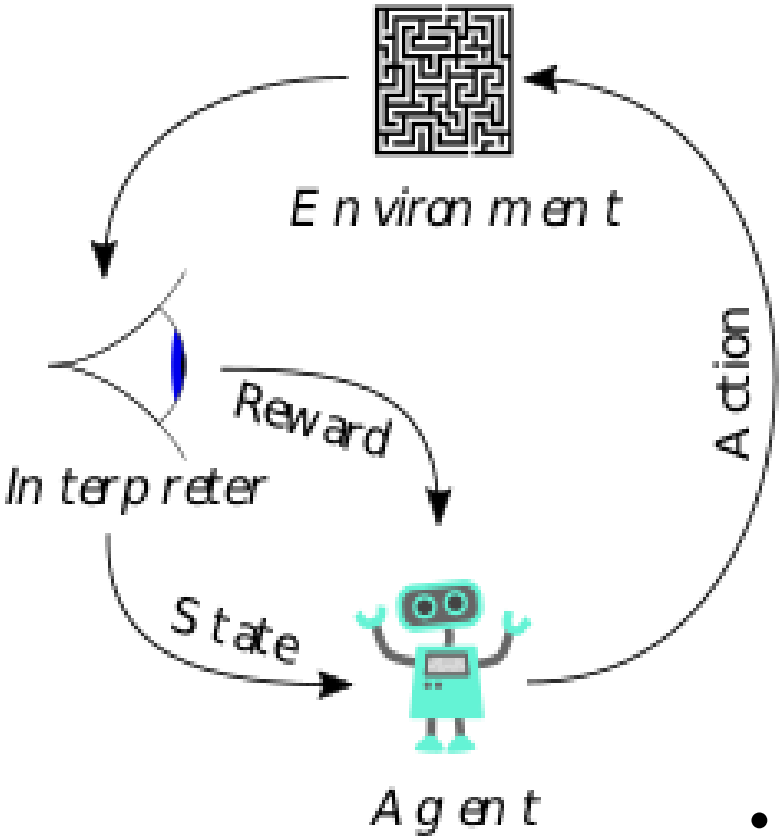
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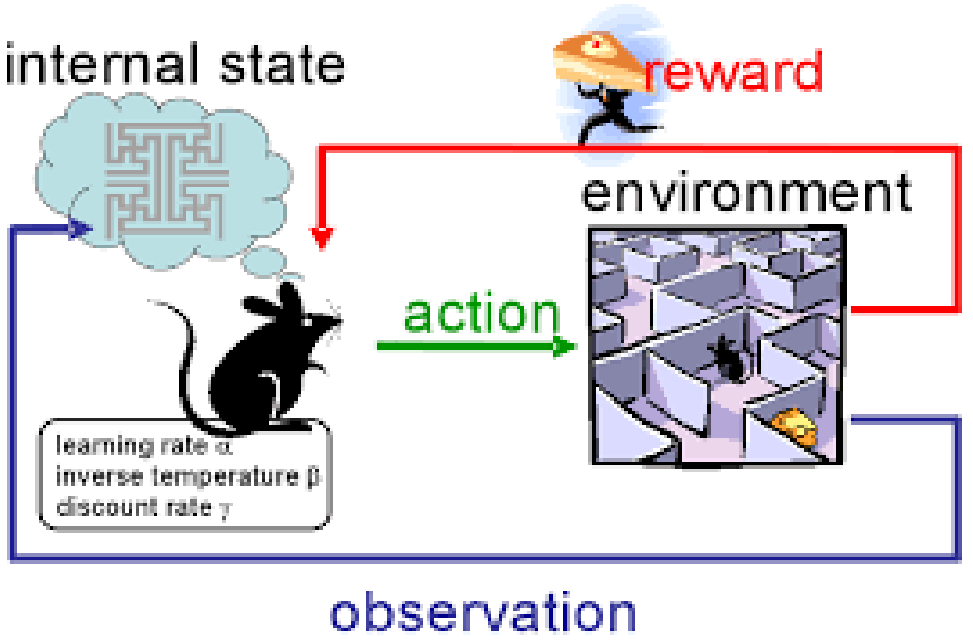
# Unsupervised Learning




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# Reinforcement Learning



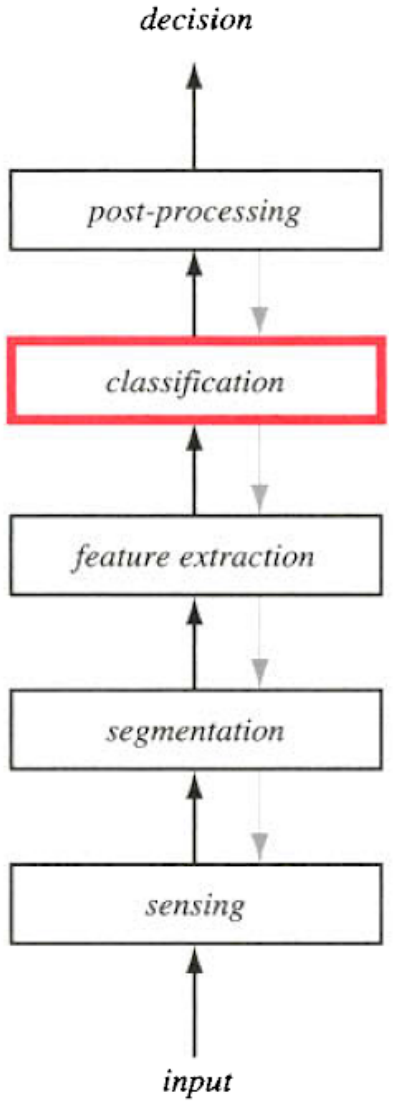




- A reinforcement learning agent interacts with its environment in discrete time steps.

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# Pattern Recognition Process

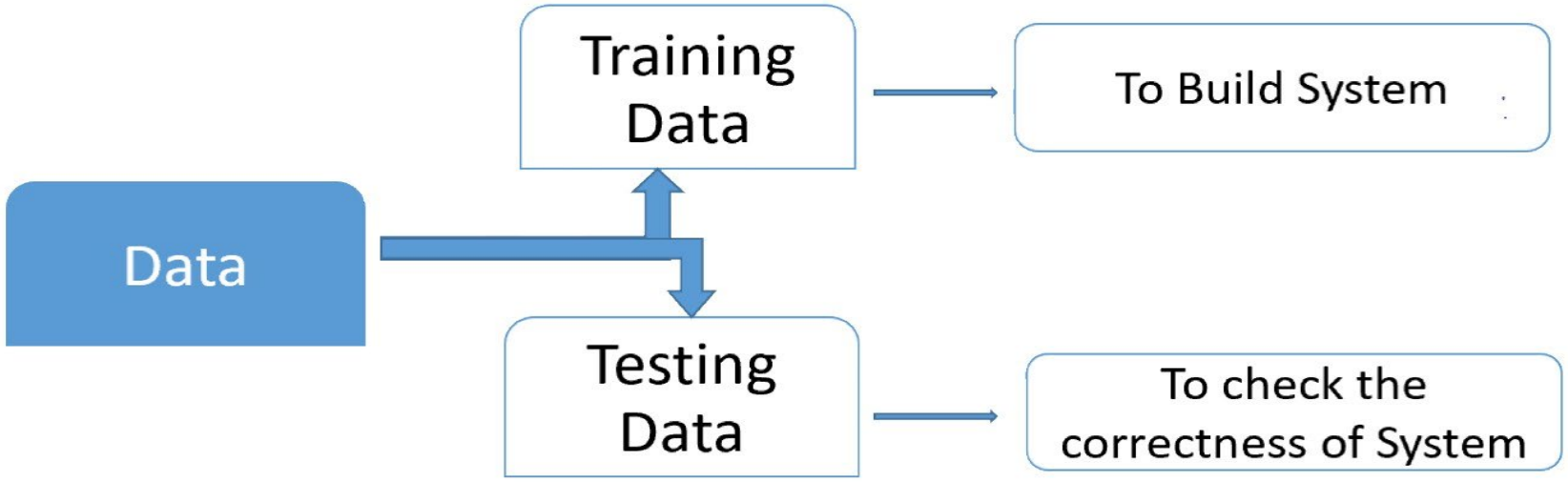
- **Data acquisition and sensing:**
  - Measurements of physical variables.
  - Important issues: bandwidth, resolution , etc.
- **Pre-processing:**
  - Removal of noise in data.
  - Isolation of patterns of interest from the background.
- **Feature extraction:**
  - Finding a new representation in terms of features.
- **Classification/Clustering**
  - Using features to learn models for different tasks.
- **Post-processing**
  - Evaluation of confidence in decisions.




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# Training and Testing Models in PR

- **Learning (training)**: Learn a model using the **training data**
- **Testing**: Test the model using unseen **test data** to assess the model accuracy



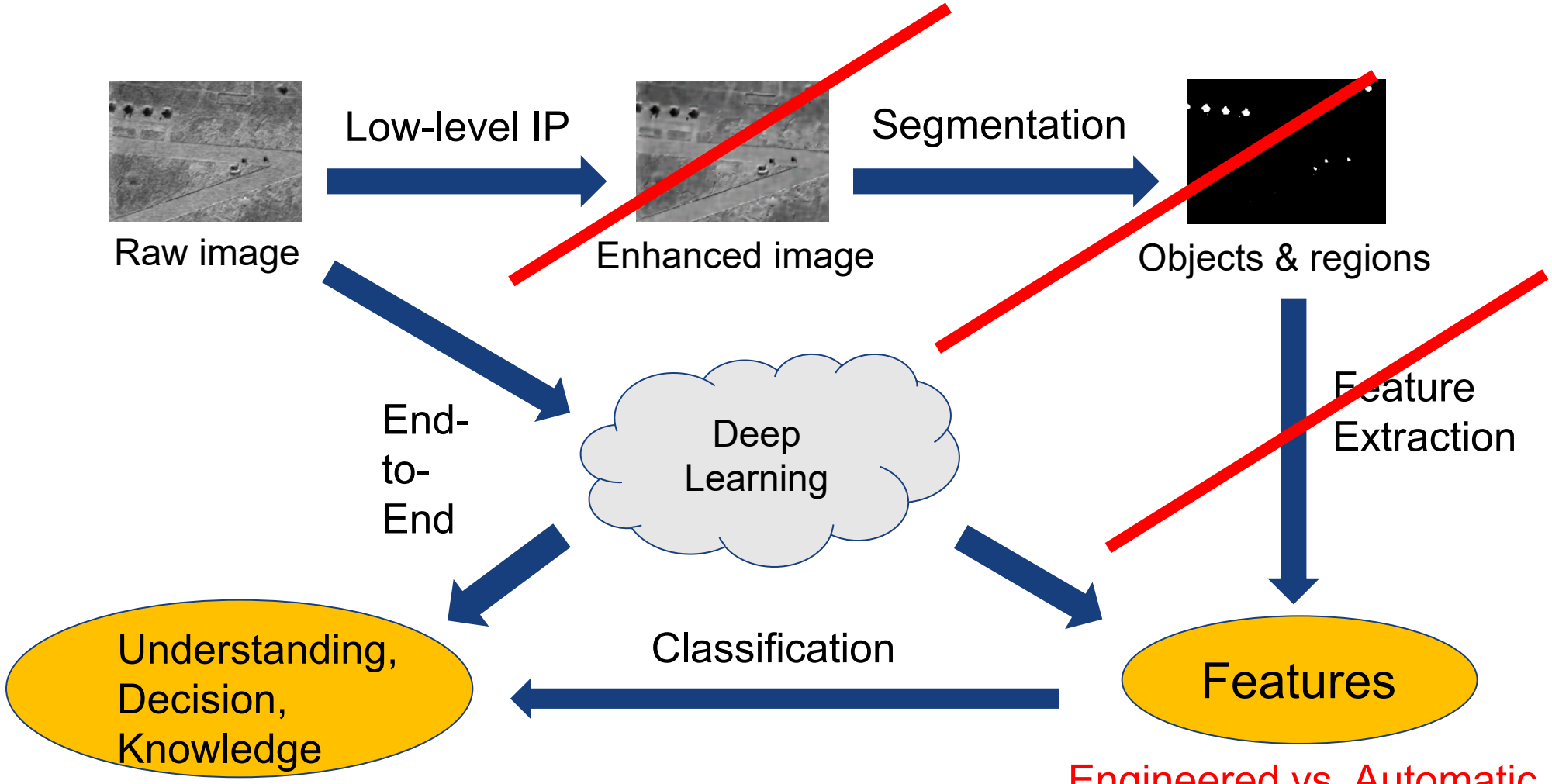

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# Machine Learning Techniques/Algorithms for PR

- Linear regression / Logistic regression
- Decision tree (C5.0, CART, …)
- Naïve Bayes, Bayesian Network
- Neural network (MLP, Radial Basis, SOM, …)
- Deep Learning (CNN, LSTM, GAN, VAE, Transformer, …)
- Support Vector Machine
- Memory-based reasoning (K-NN)
- K-Means
- Hierarchical Clustering
- ……

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# Deep Learning Paradigm




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# Applications of Pattern Recognition

- Handwritten digit/letter recognition
- Biometrics: voice, iris, fingerprint, face, and gait recognition
- Speech recognition
- Smell recognition (e-nose, sensor networks)
- Defect detection in chip manufacturing
- Interpreting DNA sequences
- Medical diagnosis
- Terrorist Detection
- Credit Fraud Detection
- Credit Applications
- … …

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# Multimodal Pattern Recognition

- **Multiple Data Types:** Processing and understanding patterns across different modalities like text, images, audio, video, and sensor data simultaneously
- **Cross-Modal Learning:** Discovering relationships and correlations between different types of data to improve recognition accuracy
- **Unified Understanding:** Creating comprehensive models that can interpret complex real-world scenarios by integrating multiple sensory inputs (e.g., we use sight + sound to recognize a person)

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# Multimodal Pattern Recognition: Applications, Challenges and Benefits

- **Applications:**
  - Healthcare: Diagnosing using patient records + medical imaging
  - Autonomous Vehicles: Combining LiDAR, camera, radar
  - Human-Computer Interaction: Emotion detection using facial + voice analysis
  - Surveillance & Security: Video + audio analysis
- **Challenges:**
  - Fusion strategies (early vs late)
  - Synchronization of modalities
  - Missing or noisy data
  - Scalability and computational complexity
- **Benefits:**
  - More robust and accurate recognition
  - Reduces ambiguity present in single-modal systems
  - Enables context-aware AI systems

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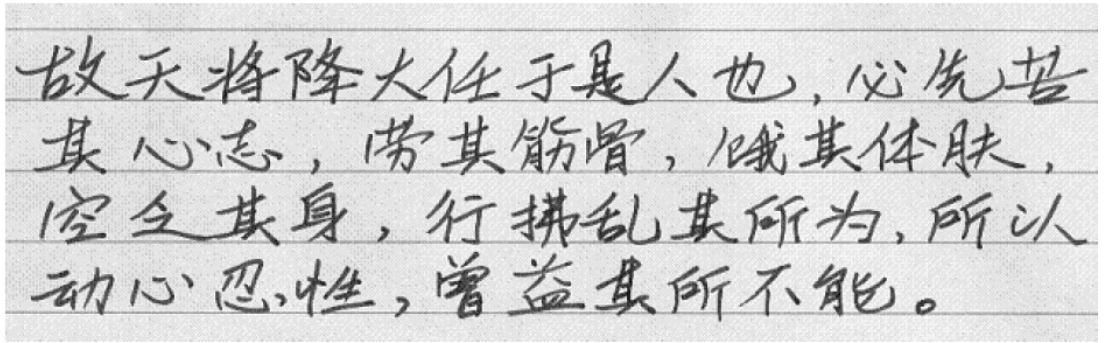
# Generative AI for Multimodal Pattern Recognition

- Generative AI enhances traditional pattern recognition by not only identifying patterns but also creating new data that follows learned patterns, enabling data augmentation and synthetic sample generation.
- Generates one modality from another (e.g., image from text, text from image).
- Enables richer pattern understanding by leveraging cross-modal generation and alignment.
- Supports tasks like image captioning, visual question answering, multimodal summarization.

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# Applications of Pattern Recognition (cont.)

- **Hand-Written Character Recognition**



(a) Handwriting

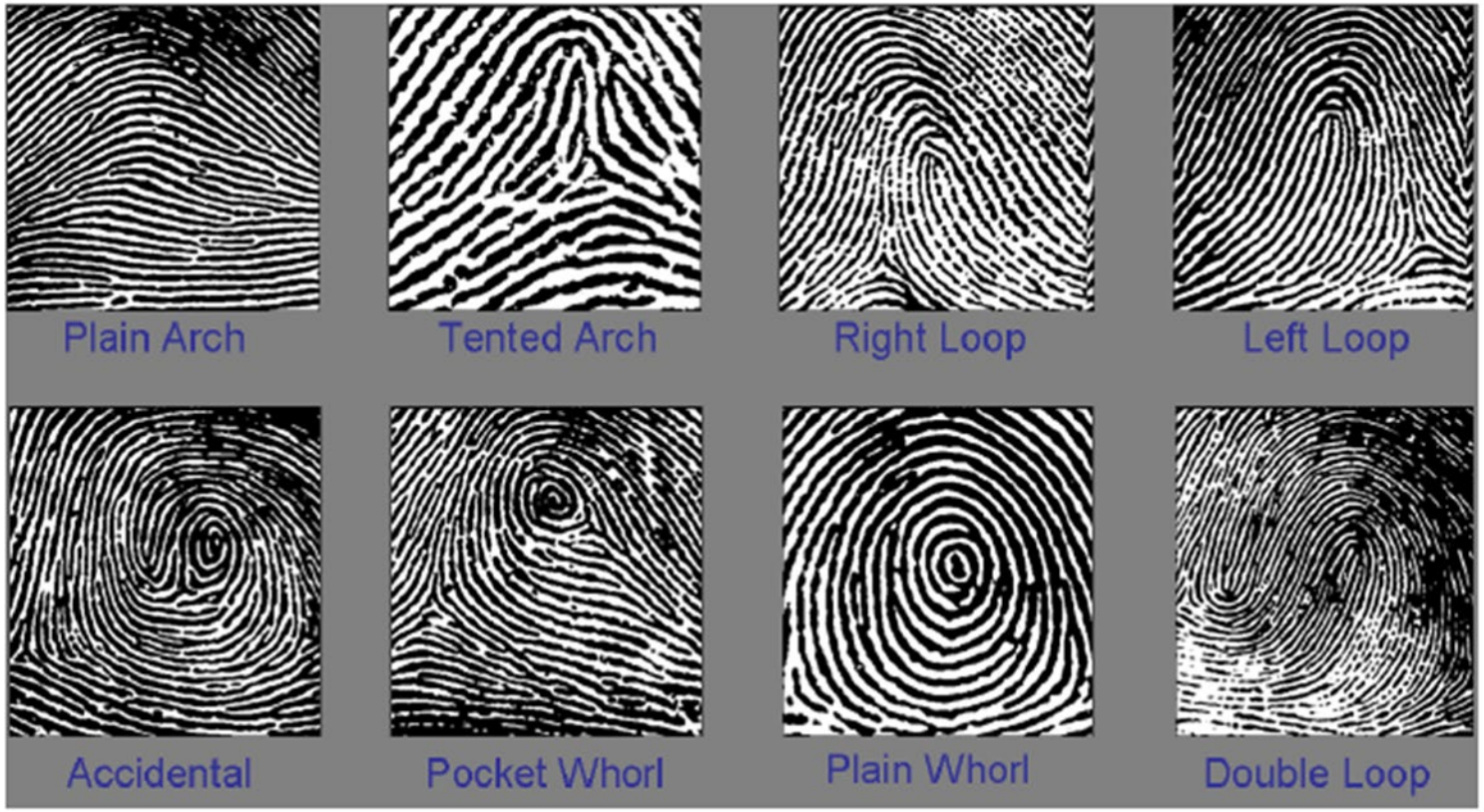
故天将降大任于是人也，必先苦
其心志，劳其筋骨，饿其体肤，
空乏其身，行拂乱其所为，所以
动心忍性，曾益其所不能。

(b) Corresponding Machine Print

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# Applications of Pattern Recognition (cont.)

- **Fingerprint Classification**

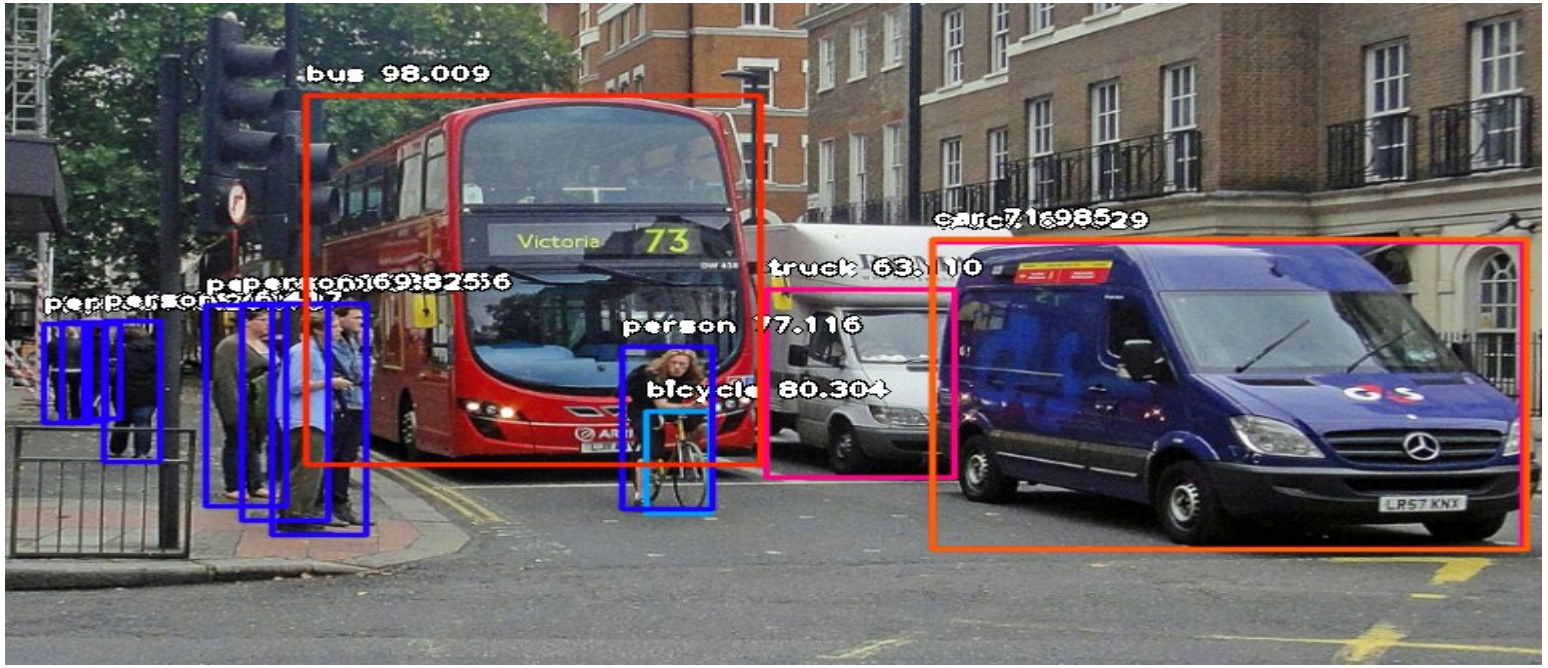



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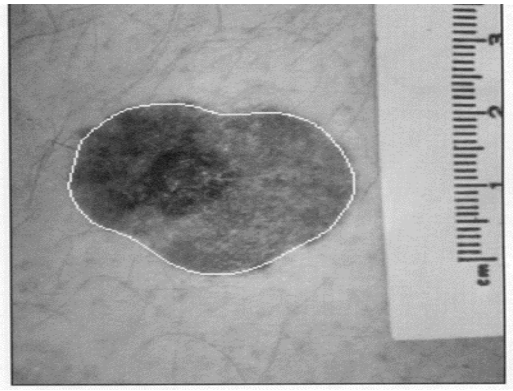
# Applications of Pattern Recognition (cont.)



Autonomous Systems




Object Detection



Medical Applications - Skin cancer detection

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